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Spatio-Temporal Machine Learning Approaches for Bus Travel Time Prediction

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Abstract

Predicting bus travel times is critical for enhancing passenger satisfaction, optimizing route planning, and improving overall system efficiency. This study explores spatio-temporal modeling techniques to predict travel times for a bus route in Chennai, India, leveraging spatial dependencies across route sections and temporal variations in travel conditions. Machine learning algorithms, including Weighted K-Nearest Neighbors (KNN) and clustering-based approaches, were compared with deep learning methods like Recurrent Neural Networks (RNN). The models incorporated handcrafted spatio-temporal features and real-time data to improve prediction accuracy. The results reveal that machine learning models with spatio-temporal feature engineering performed competitively with deep learning models utilizing real-time data. The study presents the strengths and limitations of each approach, offering insights into effective spatio-temporal modeling for bus travel time prediction.

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Keywords: travel time prediction; KNN; clustering; RNN; spatio-temporal modeling

1. Introduction

Efficient and reliable public transportation systems play a pivotal role in urban mobility, shaping the daily lives of millions of people worldwide. Among the various modes of public transit, buses stand out as a critical component, offering flexibility, accessibility, and affordability to passengers. However, the effectiveness of bus services heavily relies on accurate travel time predictions, as they directly impact passenger satisfaction, route planning, and overall system efficiency. In recent years, the integration of advanced data analytics and machine learning techniques has revolutionized the approach of transportation agencies to predicting bus travel times. This paper focuses on spatio-

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temporal modeling to uncover dynamic patterns in travel times by leveraging real-time data and machine learning approaches. By integrating spatial dependencies across route sections and temporal variations in traffic conditions, this study aims to improve the accuracy of travel time predictions. The chosen methodologies include weighted K-Nearest Neighbors (KNN), clustering-based methods, and Recurrent Neural Network (RNN) based approaches. Through a comprehensive evaluation, this study highlights the strengths and limitations of these methods for bus travel time prediction. The primary objective is to assess how effectively the selected approaches utilize spatio-temporal modeling to estimate travel times. This paper is divided into five sections. In Section 2, the literature review of various bus travel time prediction methods is discussed. Section 3 discusses data collection and the three proposed methods for bus travel time prediction - KNN based, clustering based, and RNN based methods. Section 4 discusses the implementation and results of the proposed methods. Finally, Section 5 presents a summary of the work with conclusions drawn.

2. Literature Review

Over the years, various data-driven methodologies have emerged to forecast bus travel times, ranging from conventional techniques to more sophisticated machine learning algorithms. Conventional methods, such as historical data-based approaches [1], statistical models such as regression models [2, 3, 4] and Auto-Regressive Integrated Moving Average (ARIMA) [5, 6], have been extensively employed for bus travel time prediction. These models rely on historical travel time data and incorporate factors such as time of day, day of the week, and section of the route to predict future travel durations. The introduction of machine learning techniques propelled bus travel time prediction to new heights, yielding better prediction accuracies as compared to their statistical counterparts. Various studies have utilised classification methods like K-Nearest Neighbors (KNN) that predicts travel times based on closest historical instances to a current instance in a feature set [7] and Support Vector Machines (SVM) [8, 9, 10], which capitalize on historical data patterns to make accurate predictions. SVM has also been used in high-variable data situations to learn complex relationships between various features [11].

Artificial Neural Networks (ANN) based methods harness neural networks to discern patterns from extensive training datasets and make predictions [12, 13, 14, 15]. The ANN model typically consists of the input layer, hidden layer, and the output layer which is used to identify significant variables and assign weights to the variables in the prediction of travel times. These approaches often surpass traditional statistical models by capturing intricate relationships among various factors influencing travel times. Unsupervised machine learning algorithms, such as clustering, have been utilized to identify natural groupings within the data and conduct trend prediction in travel times using pattern sequence-based forecasting (PSF) techniques [16, 17]. Furthermore, an iterative joint clustering and prediction approach has been deployed to accurately predict spatio-temporal patterns in travel times by linking the clustering module with the prediction module and training a prediction model on each cluster [18]. Hierarchical clustering has been implemented for the categorization of bus driver behaviour to then be used along with the bootstrap method for accurate travel time predictions [19]. Density Based Clustering is another technique used for pattern-based predictions based on the concentration of datapoints and the separation of these areas [20].

A few studies have been reported to construct dynamic travel time prediction models using some of the above-mentioned machine learning techniques [20, 21, 22]. However, within these methodologies, a significant gap still persists in the extraction of dynamic travel time patterns using spatio-temporal modeling and the utilization of real-time travel time data to achieve better predictions. This study endeavors to address this gap by refining and optimizing existing machine learning-based algorithms to accommodate spatiotemporal variances and facilitate dynamic predictions based on real-time data, particularly in mixed traffic conditions.

3. Methodology

This section outlines the methodology for predicting bus travel times. Beginning with data collection and preprocessing to enhance quality, the study introduces three distinct approaches: Weighted KNN, Clustering-based methods, and RNN-based methods. Weighted KNN extracts similar travel time patterns from historical data for accurate predictions. Clustering-based methods identify hidden similarities across spatiotemporal points, enhancing prediction robustness. RNN-based methods capture temporal and spatial dependencies in sequential data for accommodating

dynamic travel time patterns. Each approach is comprehensively explained, outlining the underlying principles and implementation strategies.

3.1. Data Collection

The data used in this study was collected through Global Positioning System (GPS) devices installed on buses operated by the Metropolitan Transport Corporation in Chennai, India. This study focused on the northbound 19B bus route, which spans 29.4 kilometers, connecting Kelambakkam, a suburban area, to Saidapet, a prominent commercial hub. GPS data was captured at 5-second intervals from the GPS units installed on the buses, resulting in a total of 1,231 trips being recorded over a 34-day period. The collected data included information such as date, timestamp, latitude, and longitude of the bus's position. The route was partitioned into 500-meter segments, yielding 56 distinct sections. Travel times for each of these sections were computed through interpolation techniques. The distance between consecutive GPS points was calculated using the Haversine formula. The final dataset included tripwise information such as date, trip number, start time along with section wise travel times for each trip.

3.2. Weighted KNN

Weighted KNN is a machine learning algorithm used for classification and regression tasks. In this context, weighted KNN is used for regression by identifying historical instances of similar travel conditions (similar travel times in previous sections, previous trips, etc.) and averaging them to estimate the travel time for the upcoming section. Weighted KNN assigns different weights to each neighbor based on their proximity to the query point, with closer neighbors receiving higher weights. The equation of weighted KNN is given in Equation 1,

$$\hat{y}(x_q) = \frac{\sum_{i=1}^{K} \frac{1}{d(x_q,x_i)} y_i}{\sum_{i=1}^{K} \frac{1}{d(x_q,x_i)}}, \tag{1}$$

where $\hat{y}(x_q)$ is the predicted value for the query point x_q, K is the number of nearest neighbors considered, $d(x_q, x_i)$ is the euclidean distance between the query point x_q and the i-th nearest neighbor x_i, and y_i is the target variable value associated with the i-th nearest neighbor.

Three distinct sets of input features (x_q) were examined for the implementation of weighted KNN. The initial approach entailed a three-dimensional input vector (tt_{n-3}, tt_{n-2}, tt_{n-1}), where the travel times of the preceding three sections comprised the input features. This feature vector facilitates weighted KNN in identifying segments of trips with similar trends of travel times within a trip. The second approach incorporated a four-dimensional input vector (tt_{n-3} , tt_{n-2} , tt_{n-1} , et_n), encompassing the travel times of the preceding three sections along with the entry time at the subsequent section as the fourth feature. This feature vector aids KNN in identifying segments of trips with similar trends of travel times, while also facilitating the identification of better neighbors by incorporating the time of the trip as an additional feature. Finally, the third approach adopted a three-dimensional input vector ($tt_{n-1,t}$, $tt_{n,t-1}$, $tt_{n,d-1}$), where the first dimension represented the travel time in the prior section, the second dimension denoted the travel time of the current section during the preceding trip, and the third dimension denoted the travel time of the current section at the same time on the previous day. This feature vector assists in the identification of better neighbors across both space (sections) and time (trips). The target value $\hat{y}(x_q)$ remained consistent across all approaches, which is the predicted travel time in the current section (tt_n) for the current trip.

3.3. Clustering based prediction

Clustering is an unsupervised machine learning algorithm used to identify inherent structures or patterns within datasets. Clustering methods partition data points into distinct groups or clusters based on their similarities to discover hidden relationships and insights within the dataset. In this study, KMeans clustering is performed on the obtained section-wise travel times across all the trips. The optimal number of clusters was identified to be four, using the elbow method. The travel times are accordingly grouped into four clusters - *low*, *medium*, *high* and *very high*. Each data point is mapped to its corresponding cluster and a cluster distribution matrix M is created based on the frequency of data points in each cluster [*low*, *medium*, *high*, *very high*] for each (hour, section) combination as shown in Figure

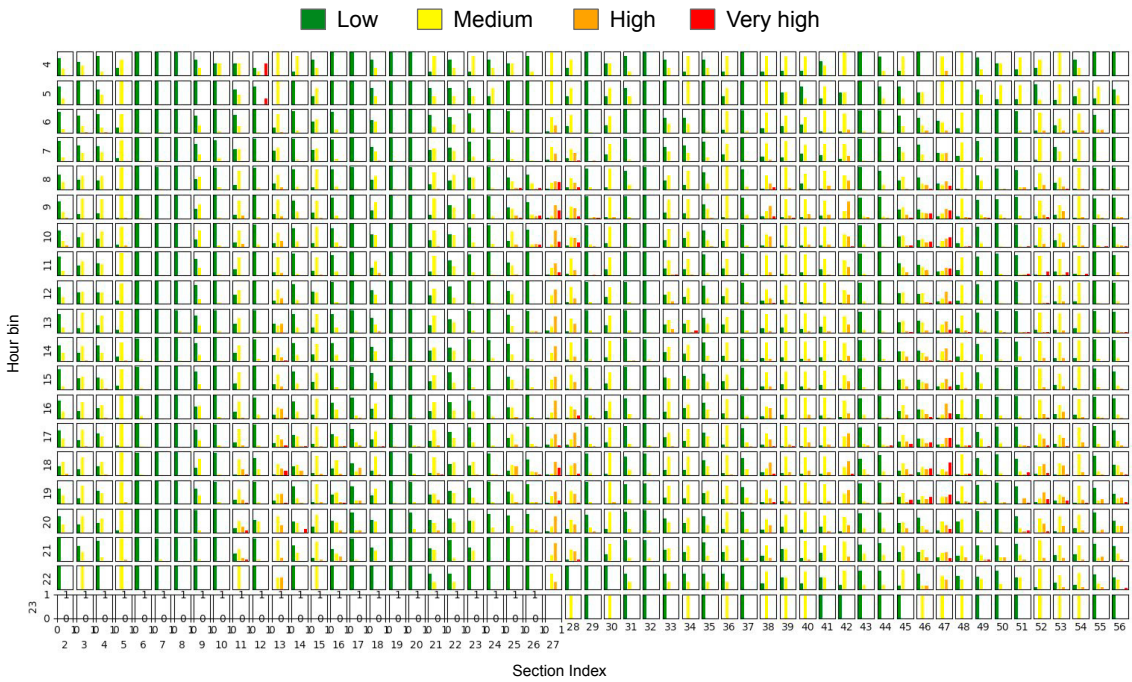

Fig. 1. Plot of cluster distribution matrix M.

1. It shows the distribution of data points among the four clusters for each (hour, section). This matrix is used as the reference for the clustering based prediction methods. Three distinct clustering-based prediction methods were investigated in this study. Two of them utilize the same input features and target values, denoted as $\{x, y\}$, where $x : (h_n, n)$ and $y : tt_n$. Here, h_n represents the hour value of entry into section n, and tt_n signifies the travel time for section n. The differentiation among these two methods lies in their treatment of the distribution for the input (h_n, n) to derive the predicted travel time tt_n.

The first approach involves selecting the centroid of the cluster with the highest frequency corresponding to the (hour, section) distribution as the predicted travel time. This method, detailed in equation 2, identifies the cluster with the maximum frequency for each (hour, section) combination and using the centroid (μ) of that cluster as the predicted travel time tt_n.

$$tt_n = \mu(\underset{c \in \{\text{low,medium,high,very high}\}}{\arg\max} M((h_n, n), c)) \tag{2}$$

$$tt_n = \frac{\sum f(c_i) \cdot \mu(c_i)}{\sum f(c_i)} \quad \forall c_i \in M((h_n, n), c_i) \tag{3}$$

The second approach involves calculating the weighted average of centroids of all clusters for the corresponding (hour, section) cell. Each cluster's centroid is weighted by the frequency of the cluster counts for that (hour, section) combination. This method, as given in equation 3, assumes that each (hour, section) cell is characterized by the distribution of clusters in that cell. The predicted travel time tt_n is taken as the weighted average of centroids of all the clusters.

$$tt_{t,n} = \begin{cases} \frac{1}{K} \sum_{i=1}^{K} tt_{i,n}, & \text{if correlation}(n, n-1) > T \\ \mu(\underset{c \in \{\text{low,medium,high,very high}\}}{\arg\max} M((h_n, n), c)), & \text{otherwise} \end{cases} \tag{4}$$

The third approach improves upon the previous methods by fine-tuning the clustering based predictions, as detailed in Equation 4. It aims to capture the variation in travel times within the same cluster using KNN. For each trip, the

algorithm checks if the correlation between the current and previous section exceeds a threshold T. If the correlation is high, the cluster c corresponding to the previous section's actual travel time $tt_{t,n-1}$ is identified, and the K nearest neighbors from past trips within that cluster are retrieved. Their corresponding travel times for the current section $tt_{i,n}$ are averaged to predict the travel time for current trip $tt_{t,n}$. If the correlation is low or there are no neighbors available for a specific $tt_{t,n-1}$, the prediction falls back to the maximum frequency centroid method.

3.4. RNN

RNNs are a class of artificial neural networks particularly adept at modeling sequential data and time-series dependencies. RNNs possess recurrent connections that allow them to retain and utilize information from previous time steps, making them well-suited for tasks involving temporal dynamics and sequential patterns. In the domain of bus travel time prediction, RNNs offer a powerful framework for capturing the inherent temporal and spatial dependencies present in historical travel time data. By processing sequential input data over time, RNNs can learn to discern complex patterns and trends that influence bus travel times, thereby enabling accurate predictions for future time points.

Three approaches were adopted in terms of input data for the RNN model. The first approach consists of pairs $\{x, y\}$, where x is an T dimensional feature vector representing the travel times of section n in the previous T trips, and y is the target travel time value for section n in the current trip. This input feature vector encapsulates the sequential pattern of travel times for a particular section across trips, as a time series. The second approach consists of pairs $\{x, y\}$, where x is an S dimensional feature vector representing the travel times in the previous S sections for the current trip, and y is the target travel time value for section n in the current trip, as a space series. This input feature vector captures the sequential pattern of travel times for the current trip across previous sections. The third approach predicts the target travel time value y by taking the average of travel time predictions from both time series and space series approaches.

4. Implementation and Results

This section discusses the implementation details and presents the results of the study on the proposed methods. It begins by discussing the dataset utilized in the experiments, followed by a description of the hyper-parameter tuning process for each method. Subsequently, an evaluation of the performance of the methods is provided, comparing their accuracy and robustness.

4.1. Dataset

The dataset utilized in this study comprised 34 days of travel time data, encompassing a total of 1,231 trips. Section 1 was excluded from the analysis due to consistently high and unreliable travel times caused by delays at the source bus depot. Across all sections and trips, the mean travel time for a 500-meter segment was 65.20 seconds, corresponding to an average speed of 27.61 km/h. Travel times exhibited a wide range, with the minimum recorded at 30 seconds (equivalent to an average speed of 60 km/h) and the maximum at 1,039 seconds (equivalent to an average speed of 1.73 km/h). Most sections demonstrated relatively low travel times, largely due to a significant portion of the route being an expressway connecting the city to a suburban area. Consequently, some sections (6, 7, 8, 19, 20, 24, 31, 32, 37, 43, and 44) displayed minimal variance in travel times. In contrast, certain sections (13, 27, and 47) exhibited high variance and a notable presence of outliers at the upper end of the travel time spectrum, primarily attributable to the presence of signalized intersections within these segments. For the purpose of this study, the dataset was partitioned into training, validation, and test sets in chronological order, given the predictive nature of the task. Accordingly, data from the initial 20 days (comprising 600 trips) were allocated for training, followed by the subsequent 7 days (comprising 326 trips) for validation, and finally, the last 7 days (comprising 305 trips) for testing. The methods compared in this study need only the most recent three weeks of data for training, as this is sufficient to capture recent travel time trends and allows for easy implementation in any location.

4.2. Hyper-parameter tuning

The optimization of machine learning models depends significantly upon the hyper-parameter tuning process. The most effective hyper-parameter configurations for all three methods is discussed below.

Weighted KNN: Weighted KNN has three parameters: the number of neighbors considered for predictions, the type of weights, and the distance metric used for calculating distances between points. For all three KNN approaches, the chosen weight type was distance-based, where farther neighbors contribute less and closer neighbors contribute more. The Euclidean distance metric was selected to measure distances between points. However, determining the optimal number of neighbors involved employing the elbow method. The Mean Absolute Percentage Error (MAPE) was calculated on the validation set and plotted against different values of K. The optimal K was identified as the point beyond which there was no significant reduction in MAPE. The optimal number of neighbors for Method 1, Method 2, and Method 3 are 40, 20, and 50 respectively.

Clustering: KMeans clustering was performed to determine the optimal number of clusters for the given dataset. The optimal value of K was identified using the elbow method, which involves plotting inertias (the sum of squared distances between each point and its cluster centroid) against the number of clusters. After analysis, the optimal number of clusters was determined to be $K = 4$, as no significant reduction in inertia values was observed beyond this point. Based on the clustering results, four clusters were defined: *low*, *medium*, *high*, and *very high* travel times. The centroids of these clusters were calculated as 42.14 seconds for *low*, 74.96 seconds for *medium*, 154.06 seconds for *high*, and 345.70 seconds for *very high*. The *low* cluster accounted for the majority of the data points, comprising 55.67% of the dataset, while the *very high* cluster represented the smallest proportion, consisting of only 1.29% of the data points. The *medium* and *high* clusters comprised 36.77% and 6.27% of the data points, respectively.

RNN: RNN models consist of several hyperparameters, including the learning rate, number of steps, number of hidden units, and batch size. These hyperparameters significantly impact the model's performance and require careful tuning. Both the time series and space series models were trained and evaluated across a range of values for each hyperparameter. The optimal values were determined based on the error observed on the validation set. For the time series model, the optimal hyperparameter values were found to be a learning rate of 0.0001, 512 hidden units, a batch size of 256, and 5 previous trips. Similarly, for the space series model, the optimal hyperparameter values were a learning rate of 0.0001, 128 hidden units, a batch size of 128, and 10 previous sections.

4.3. Evaluation

The evaluation assesses the performance of each method in accurately predicting bus travel times. The performances of all the methods are evaluated for weekdays and are presented in Table 1. Methods 1 and 2 of the weighted KNN demonstrate poor performance throughout the week, indicating that their feature vectors are ineffective in identifying appropriate neighbors. In contrast, Method 3 performs significantly better due to a hand-crafted feature vector that facilitates effective neighbor identification. Among clustering-based methods, the maximum frequency centroid and clustering+KNN approaches perform comparably and outperform the weighted average of centroids. For RNN models, the time series approach achieves the best results, effectively capturing the temporal variation within each section. Conversely, the space series approach underperforms, as it fails to account for the inherent temporal variation embedded in the spatial variation of sections during the bus journey. Overall, the clustering+KNN model and the time series RNN model deliver comparable results on all weekdays, with similar MAPE values.

Clustering+KNN and time series RNN models were selected for further analysis due to their superior performance compared to other methods. Table 2 presents the MAPE across all trips for selected low and high variance sections along the bus route. Clustering+KNN outperforms time series RNN in low variance sections, significantly reducing MAPE in section 7 by selecting appropriate neighbors within the same cluster. Conversely, time series RNN excels in high variance sections such as 27 and 47, with a notable improvement in MAPE for section 47. This highlights its ability to effectively capture high travel time variations using previous trip data.

Analyzing performance across different times of the day, as shown in Table 3, Clustering+KNN performs better during morning and evening peak hours. This demonstrates its capability to leverage spatio-temporal clustering to model travel time distributions and provide accurate predictions. In contrast, the time series RNN model performs slightly better during off-peak hours.

Table 1. MAPE across all sections and trips for weekdays.

Type of Model	Input Features / Method	MAPE across all sections and trips				
		Mon	Tue	Wed	Thu	Fri
Weighted KNN	$(tt_{n-3}, tt_{n-2}, tt_{n-1})$	40.73	40.19	39.37	38.57	39.78
	$(tt_{n-3}, tt_{n-2}, tt_{n-1}, et_n)$	42.08	41.16	41.30	38.74	38.98
	$(tt_{n-1,t}, tt_{n,t-1}, tt_{n,d-1})$	24.02	24.86	24.82	23.59	24.63
Clustering Based	Maximum frequency centroid	24.55	22.86	22.31	22.46	23.19
	Weighted average of centroids	28.91	25.21	24.48	23.12	23.31
	Clustering+KNN	23.70	**22.84**	21.86	**22.16**	**22.55**
RNN	Time series: $(tt_{n,t-5}, tt_{n,t-4}, tt_{n,t-3}, tt_{n,t-2}, tt_{n,t-1})$	**23.23**	23.76	**21.31**	22.44	23.55
	Space series: $(tt_{n-10,t}, tt_{n-9,t}, \ldots, tt_{n-1,t})$	32.77	35.67	32.54	34.18	35.22
	Time and space series	26.20	27.72	24.99	26.01	27.56

Table 2. MAPE across all trips for different sections.

Type of Model	Input Features / Method	MAPE across all trips					
		Low variance sections			High variance sections		
		7	24	37	13	27	47
Clustering Based	Clustering+KNN	**8.01**	**13.91**	**16.52**	**27.00**	36.79	83.64
RNN	Time series: $(tt_{n,t-5}, tt_{n,t-4}, tt_{n,t-3}, tt_{n,t-2}, tt_{n,t-1})$	15.79	17.65	21.37	32.39	**28.44**	**48.24**

Table 3. MAPE for section 13 across different times of a day.

Type of Model	Input Features / Method	MAPE for section 13				
		Early morning 04:00 - 08:00	Morning 08:00 - 12:00	Afternoon 12:00 - 16:00	Evening 16:00 - 20:00	Night 20:00 - 00:00
Clustering Based	Clustering+KNN	12.15	**19.96**	27.53	**46.99**	**17.42**
RNN	Time series	**10.19**	30.92	**25.28**	61.78	30.70

These findings underscore the potential of traditional machine learning methods to handle complex patterns in bus travel time prediction, particularly when the feature space is carefully engineered to account for spatial and temporal variations.

5. Conclusion

In conclusion, this study evaluated three spatio-temporal modeling approaches for predicting bus travel times: Weighted KNN, clustering-based techniques, and RNNs. The findings underscore the importance of leveraging spatio-temporal dependencies—such as the sequential progression of travel times and variations across route sections—to enhance prediction accuracy and reliability. Notably, machine learning models with spatio-temporal feature engineering performed comparably to deep learning models, suggesting that careful design of handcrafted features can bridge the gap between traditional and computationally intensive advanced techniques. While travel time inherently captures traffic and weather conditions, future work could integrate real-time traffic and weather data as additional features to further enhance predictions.

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